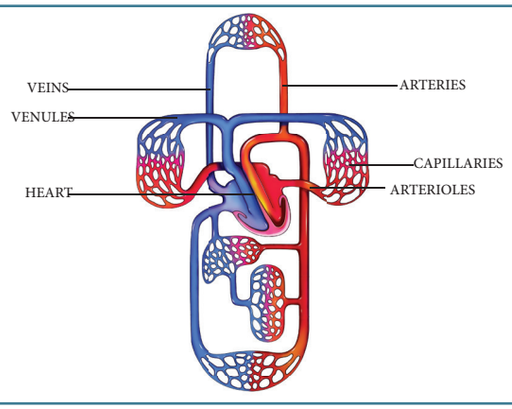
## BLOOD CIRCULATION

Circulation can be roughly divided into pulmonary and systemic circulation.

1. Pulmonary circulation
2. Systemic circulation

## PULMONARY CIRCULATION

Deoxygenated blood from all parts of the body reaches the right atrium through two major veins the superior and inferior vena cava. From the right atrium this blood reaches the right ventricle. From the right ventricle, blood flows into the pulmonary artery through which it is supplied to both lungs. In the lungs, the blood gets oxygenated.




## SYSTEMIC CIRCULATION

The oxygenated blood from the lungs enters the left atrium through 4 pulmonary veins. From the left atrium the blood enters the left ventricle. From the left ventricle, through aorta, and its branches. This oxygenated blood is supplied to all parts of the body.

## FUNCTIONS OF HEART

- It draws blood back from the capillaries and veins.
- It sends blood into the lungs where it is oxygenated.
- It sends blood through the aorta to all parts of the body.

## BLOOD VESSELS

There are mainly 3 types of blood vessels:

- Arteries
- Veins
- Capillaries.

## BLOOD

About 6 litres of blood continuously circulates through the heart and blood vessels in all parts of the body. It is a sticky red fluid is slightly alkaline in reaction. It made up of a liquid (Plasma) and solid (cells).

### STUDENT'S ACTIVITY

Separation of plasma and cells in a test tube at classroom.

Plasma is a pale yellow fluid consisting of:

- Water 90 %
- Salts, including sodium chloride
- Proteins (Albumin, globulin, fibrinogen)
- Nutrients, such as glucose, fats, amino acids, vitamins and minerals
- Waste products such as Urea and carbon dioxide
- Antibodies and antitoxins for resistance to germs.



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